

Coherence Calculus — Unified Clean Specification (v1)

A self-contained specification that unifies the coherence-object calculus with a coherence-centric arithmetic layer. All constructions are mathematically well-posed and ready for formalization.

0. Conventions

- Ground field: $\mathbb{R}$ unless explicitly complexified.
- Inner products are $\mathbb{R}$ -bilinear, positive-definite.
- Sections are measurable; analysis is carried out in H^1 unless noted.
- $\langle \cdot, \cdot \rangle_H$ denotes the Hilbert inner product on a space H ; $\| \cdot \|_H$ its norm.
- ∇ denotes a fixed compatible connection.

1. Geometry and Hilbert structure

Let M be a connected, oriented, compact Riemannian manifold with metric g . Let $\text{Cl}(TM, g)$ be the real Clifford bundle with the fiberwise inner product induced by g . Denote by $\Gamma(\text{Cl})$ the measurable sections.

Fix a metric- and Clifford-compatible connection ∇ on $\text{Cl}(TM, g)$. Define the Hilbert space

$$H := H^1(M, \text{Cl}(TM, g)), \quad \langle \sigma, \tau \rangle_H := \int_M (\langle \nabla \sigma, \nabla \tau \rangle + \langle \sigma, \tau \rangle) d\text{vol}_g.$$

If $\partial M \neq \emptyset$ use H_0^1 with Dirichlet boundary conditions.

2. Orthonormal coordinates and functionals

Because H is separable, fix an H -orthonormal basis (ONB) $\mathcal{B} = \{B_i\}_{i \in \mathbb{N}} \subset H$. Define continuous linear functionals

$$L_i(\sigma) := \langle \sigma, B_i \rangle_H, \quad i \in \mathbb{N}.$$

If $\sigma = \sum_i c_i B_i$ (convergent in H) then $c_i = L_i(\sigma)$.

One canonical choice sets $B_j := (\lambda_j + 1)^{-1/2} \phi_j$, where ϕ_j are eigen-sections of $\nabla^* \nabla$ with eigenvalues λ_j , yielding an H -ONB.

3. Arithmetic layer: encoding and decoding

Let $\mathbb{P}$ be the set of primes. Fix a bijection $\psi : \mathbb{P} \rightarrow \mathbb{N}$. Define the encoder $F : \mathbb{N}_{\geq 1} \rightarrow \ell^2(\mathbb{N})$ by

$$F(n)_i := v_{\psi^{-1}(i)}(n) \in \mathbb{N}, \quad \text{the exponent of the prime } \psi^{-1}(i) \text{ in } n.$$

Each $F(n)$ has finite support. Define the decoder on integer-valued sequences of finite support:

$$D(e) := \prod_{i \in \mathbb{N}} \psi^{-1}(i)^{e_i} \in \mathbb{N}_{\geq 1}.$$

Then $D \circ F = \text{id}$ and $F(ab) = F(a) + F(b)$ coordinatewise.

4. Representation sets

For $n \geq 1$ define the closed affine set

$$\text{Rep}(n) := \{ \sigma \in H : L_i(\sigma) = F(n)_i \text{ for all } i \}.$$

Nonemptiness holds because $F(n)$ has finite support, so $\sum_{i \in \text{supp} F(n)} F(n)_i B_i \in \text{Rep}(n)$. The tangent space at any point is $\ker K$ with $K : H \rightarrow \ell^2$, $K(\sigma) = (L_i(\sigma))_i$.

5. Energy and coherence embedding

Define the strictly convex, coercive quadratic energy $E(\sigma) := \frac{1}{2} \|\sigma\|_H^2$. For $n \geq 1$ define the embedding

$$\text{embed}(n) := \arg \min_{\sigma \in \text{Rep}(n)} E(\sigma).$$

Existence and uniqueness follow from strict convexity and closedness of $\text{Rep}(n)$. Writing $e := F(n)$, the unique minimizer is the finite sum

$$\text{embed}(n) = \sum_{i \in \mathbb{N}} e_i B_i.$$

6. Algebraic laws

For $a, b \geq 1$ and $k \in \mathbb{N}$: 1. $\text{embed}(1) = 0$. 2. $\text{embed}(ab) = \text{embed}(a) + \text{embed}(b)$. 3. $\text{embed}(a^k) = k \text{ embed}(a)$. These follow from F and linearity of coefficients.

7. Resonance functionals (multiplicative characters)

7.1 Positive reals

Choose parameters $\alpha = (\alpha_i)_{i \in \mathbb{N}} \in (0, \infty)^{\mathbb{N}}$ with $\theta_i := \log \alpha_i \in \ell^2$. Define

$$R_\alpha(n) := \prod_i \alpha_i^{F(n)_i} = \exp(\langle F(n), \theta \rangle_{\ell^2}) \in (0, \infty).$$

Then $R_\alpha(ab) = R_\alpha(a)R_\alpha(b)$. Gradients for learning obey

$$\frac{\partial}{\partial \theta_i} \log R_\alpha(n) = F(n)_i, \quad \nabla_\theta \log R_\alpha(n) = F(n).$$

7.2 Unit circle

If $\alpha_i \in \mathbb{U}(1)$ with $\alpha_i = e^{i\theta_i}$ and $\theta \in \ell^2$, then

$$\chi_\alpha(n) := \prod_i \alpha_i^{F(n)_i} = e^{i\langle F(n), \theta \rangle}$$

is a unitary multiplicative character. If $\alpha_i \in \{\pm 1\}$ with finite support, χ_α is a $\mathbb{Z}_2$ -valued character closed under pointwise multiplication (XOR in exponents).

8. Symmetries

Let a group G act on H by an isometric representation $\rho : G \rightarrow O(H)$ and on the index set by a permutation $\pi : G \rightarrow \text{Sym}(\mathbb{N})$ such that

$$\rho(g)B_i = B_{\pi(g)(i)} \quad \text{for all } g \in G, i \in \mathbb{N}.$$

Transport ψ to $\psi_g := \psi \circ \pi(g)^{-1}$. Then

$$\rho(g) \text{ embed } \psi(n) = \text{embed } \psi_g(n), \quad \|\text{embed}(n)\|_H \text{ is } G\text{-invariant}.$$

If π permutes only indices outside $\text{supp } F(n)$ then $\rho(g) \text{ embed}(n) = \text{embed}(n)$.

9. Decoding and exact recovery

Define the coefficient readout $\text{coeff}(\sigma) := (L_i(\sigma))_{i \in \mathbb{N}} \in \ell^2$. When $\text{coeff}(\sigma) \in \mathbb{N}^{(\mathbb{N})}$, set

$$\text{decode}(\sigma) := D(\text{coeff}(\sigma)) \in \mathbb{N}_{\geq 1}.$$

Then $\text{decode}(\text{embed}(n)) = n$. If $\tilde{\sigma} = \text{embed}(n) + \eta$ with $\eta \in H$, then

$$\|\text{coeff}(\tilde{\sigma}) - F(n)\|_{\ell^2} \leq \|\eta\|_H.$$

If each coefficient error is $< \frac{1}{2}$, rounding recovers $F(n)$ and hence n .

10. Finite truncations for computation

Let P_N be the orthogonal projection onto $\text{span}\{B_1, \dots, B_N\}$. Define

$$\text{embed}_N(n) := P_N \text{ embed}(n) = \sum_{i \leq N} F(n)_i B_i.$$

Since $F(n)$ has finite support, there exists N with $\text{embed}_N(n) = \text{embed}(n)$.

11. Optional complexification

When complex scalars are needed, use $H_{\mathbb{C}} := H \otimes_{\mathbb{R}} \mathbb{C}$ with the canonical sesquilinear inner product $\langle u \otimes z, v \otimes w \rangle = z\overline{w} \langle u, v \rangle_H$. All definitions above extend verbatim. Characters χ_{α} and symmetry actions lift unitarily.

12. Interfaces summary

- **State space:** $H = H^1(M, \text{Cl}(TM, g))$.
- **Coordinates:** $L_i(\sigma) = \langle \sigma, B_i \rangle_H$ for an H -ONB $\{B_i\}$.
- **Encoder/decoder:** $F(n) = (v_{\psi^{-1}(i)}(n))_i$, $D(e) = \prod_i \psi^{-1}(i)^{e_i}$.
- **Representation sets:** $\text{Rep}(n) = \{\sigma : L_i(\sigma) = F(n)_i \ \forall i\}$.
- **Energy:** $E(\sigma) = \frac{1}{2} \|\sigma\|_H^2$.
- **Embedding:** $\text{embed}(n) = \sum_i F(n)_i B_i$.
- **Resonance:** $R_{\alpha}(n) = e^{\langle F(n), \theta \rangle}$ with $\theta \in \ell^2$; unitary variant $\chi_{\alpha}(n) = e^{i\langle F(n), \theta \rangle}$.
- **Symmetry:** isometric G -action permuting the ONB induces equivariance of embed .
- **Stability:** coefficient error ℓ^2 -bounded by $\|\eta\|_H$; rounding recovers exact n under half-integer margins.

This unified calculus ties the geometric Hilbert setting to an arithmetic interface via prime-exponent coordinates, with linear algebra control, multiplicative characters for signal design, and symmetry-aware embeddings suitable for both proof assistants and numerical implementations.